US COVID-19 Outbreak and Progression

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1 Introduction

1.1 Data Source

- **COVID-19 data**: The New York Times GitHub repository
- **Population data**: U.S. Census Bureau and the International Database (IDB)

1.2 Dataset Overview

The COVID-19 datasets record cumulative counts of coronavirus cases and deaths from state and local government and health departments spanning **January 21, 2020 to March 23, 2023**, originally released by the New York Times. Tracking began with the first reported coronavirus case in Washington State and covers nation-level, state-level, and county-level records.

> **Limitation**: Due to shortage of testing, data may underrepresent true infection counts during the initial outbreak period.

The datasets used in this project:

File	Description
`us.csv`	National cumulative cases & deaths
`us-states.csv`	State-level cumulative cases & deaths
`us-population.xlsx`	Population estimates for 51 states (U.S. Census Bureau)
`islands-population.xlsx`	Population for 4 U.S. territories (IDB)

> *Note: The Census Bureau does not provide annual estimates for Guam, Virgin Islands, American Samoa, and Northern Mariana Islands — these are supplemented by the IDB dataset.*

1.3 Research Objectives

This project reconstructs the comprehensive trajectory of the COVID-19 pandemic in the U.S. from January 2020 through March 2023. Specifically, the analysis aims to:

1. Quantify the **aggregate public health burden** of COVID-19 in the U.S.
2. Characterize the **temporal evolution** of the pandemic, identifying seasonal wave patterns.
3. Evaluate **geographic heterogeneity** in pandemic impact across states.

1.4 Key Variables

Variable	Type	Description
`date`	Date	Date cases/deaths were announced (midnight Eastern Time)
`cases`	Integer	Confirmed + probable cases based on lab testing & exposure criteria
`deaths`	Integer	Confirmed + probable COVID-19-related deaths
`state`	Character	U.S. state or territory
`fips`	Character	Federal Information Processing Series code (2-digit state, 5-digit county)

---

2 Setup: Required Packages

```
# Uncomment to install packages if needed:
# install.packages(c("tidyverse", "magrittr", "readxl", "lubridate",
#                    "stringr", "zoo", "patchwork", "usmap", "scales", "tidytext"))

suppressPackageStartupMessages({
  library(tidyverse)   # Data manipulation & visualization (tidyr, dplyr, ggplot2, etc.)
  library(magrittr)    # Pipe operator
  library(readxl)      # Read Excel files
  library(lubridate)   # Date handling
  library(stringr)     # String manipulation
  library(zoo)         # Rolling mean calculation
  library(patchwork)   # Combine plots side-by-side
  library(usmap)       # U.S. map visualization with FIPS codes
  library(scales)      # Number formatting
  library(tidytext)    # Reorder bars within facets
})
```

---

3 Data Preparation

3.1 Import Data

```
us_df         <- read.csv("data/raw/us.csv")
states_df     <- read.csv("data/raw/us-states.csv")
us_population <- read_excel(
```

```r
  "data/raw/us-population.xlsx",
  range     = "A10:F62",
  col_names = c("state", "2020 base", "2020", "2021", "2022", "2023")
)
islands_population <- data.frame(read_xlsx("data/raw/islands-population.xlsx"))
```

```r
# Quick preview of each dataset
head(us_df)
head(states_df)
head(us_population)
head(islands_population)
```

3.2 Clean & Transform

3.2.1 Check Data Quality

```r
# Structure check
str(us_df)
```

```
## 'data.frame':    1158 obs. of  3 variables:
##  $ date  : chr  "2020-01-21" "2020-01-22" "2020-01-23" "2020-01-24" ...
##  $ cases : int  1 1 1 2 3 5 5 5 5 6 ...
##  $ deaths: int  0 0 0 0 0 0 0 0 0 0 ...
```

```r
str(states_df)
```

```
## 'data.frame':    61942 obs. of  5 variables:
##  $ date  : chr  "2020-01-21" "2020-01-22" "2020-01-23" "2020-01-24" ...
##  $ state : chr  "Washington" "Washington" "Washington" "Illinois" ...
##  $ fips  : int  53 53 53 17 53 6 17 53 4 6 ...
##  $ cases : int  1 1 1 1 1 1 1 1 1 2 ...
##  $ deaths: int  0 0 0 0 0 0 0 0 0 0 ...
```

```r
str(us_population)
```

```
## tibble [53 x 6] (S3: tbl_df/tbl/data.frame)
##  $ state    : chr [1:53] ".Alabama" ".Alaska" ".Arizona" ".Arkansas" ...
##  $ 2020 base: num [1:53] 5025369 733395 7158110 3011553 39555674 ...
##  $ 2020     : num [1:53] 5033094 733017 7187135 3014546 39521958 ...
##  $ 2021     : num [1:53] 5049196 734420 7274078 3026870 39142565 ...
##  $ 2022     : num [1:53] 5076181 734442 7377566 3047704 39142414 ...
##  $ 2023     : num [1:53] 5117673 736510 7473027 3069463 39198693 ...
```

```r
str(islands_population)
```

```
## 'data.frame':    16 obs. of  4 variables:
##  $ Name      : chr  "American Samoa" "Guam" "Northern Mariana Islands" "Virgin Islands" ...
##  $ GENC      : chr  "AS" "GU" "MP" "VI" ...
##  $ Year      : num  2020 2020 2020 2020 2021 ...
##  $ Population: chr  "47,392" "168,489" "51,851" "106,290" ...
```

```r
# Missing value check
list(
  us_df              = colSums(is.na(us_df)),
  states_df          = colSums(is.na(states_df)),
  us_population      = colSums(is.na(us_population)),
  islands_population = colSums(is.na(islands_population))
)
```

```
## $us_df
##   date  cases deaths
##      0      0      0
##
## $states_df
##   date  state   fips  cases deaths
##      0      0      0      0      0
##
## $us_population
##     state 2020 base      2020      2021      2022      2023
##         1         1         1         1         1         1
##
## $islands_population
##       Name       GENC       Year Population
##          0          0          0          0
```

3.2.2 Wrangle Population Data

```r
# Tidy the main population data: wide → long, remove leading "." from state names
us_population <- us_population %>%
  na.omit() %>%
  select(state, `2020`, `2021`, `2022`, `2023`) %>%
  mutate(state = str_remove(state, "^\\.")) %>%
  pivot_longer(cols = -state, names_to = "year", values_to = "population") %>%
  mutate(year = as.numeric(year))

# Tidy the islands population data
islands_pop_tidy <- islands_population %>%
  select(Name, Year, Population) %>%
  rename(state = Name, year = Year, population = Population) %>%
  mutate(population = parse_number(population))

# Combine all states + islands into one population lookup table
us_pop_combined <- bind_rows(us_population, islands_pop_tidy)
```

3.2.3 Join Population into COVID Data

```r
# Join state-level population
states_df <- states_df %>%
  mutate(year = year(date)) %>%
  left_join(us_pop_combined, by = c("state", "year"))
```

```r
# Aggregate U.S.-level population by year and join
us_pop_by_year <- us_pop_combined %>%
  group_by(year) %>%
  summarise(population = sum(population), .groups = "drop")

us_df <- us_df %>%
  mutate(year = year(date)) %>%
  left_join(us_pop_by_year, by = "year")
```

```r
# Check for states with missing population after join (indicates unmatched keys)
missing_pop_states <- states_df %>%
  filter(is.na(population)) %>%
  distinct(state, year)

if (nrow(missing_pop_states) == 0) {
  message("Population join validation passed: no missing values.")
} else {
  warning("Missing population for the following state-year combinations:")
  print(missing_pop_states)
}
```

3.2.4 Fix Column Types

```r
# Convert date columns: character → Date (ensures chronological sorting)
us_df$date     <- as.Date(us_df$date)
states_df$date <- as.Date(states_df$date)

# Pad FIPS codes with leading zeros (e.g., 6037 → "06037")
states_df <- states_df %>%
  mutate(fips = str_pad(fips, width = 2, side = "left", pad = "0"))
```

```r
# Verify final data types
str(us_df)
```

```
## 'data.frame':    1158 obs. of  5 variables:
##  $ date      : Date, format: "2020-01-21" "2020-01-22" ...
##  $ cases     : int  1 1 1 2 3 5 5 5 5 6 ...
##  $ deaths    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ year      : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ population: num  3.35e+08 3.35e+08 3.35e+08 3.35e+08 3.35e+08 ...
```

```r
str(states_df)
```

```
## 'data.frame':    61942 obs. of  7 variables:
##  $ date      : Date, format: "2020-01-21" "2020-01-22" ...
##  $ state     : chr  "Washington" "Washington" "Washington" "Illinois" ...
##  $ fips      : chr  "53" "53" "53" "17" ...
##  $ cases     : int  1 1 1 1 1 1 1 1 1 2 ...
##  $ deaths    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ year      : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ population: num  7727209 7727209 7727209 12799088 7727209 ...
```

3.3 Tidy Data

3.3.1 Cumulative Cases & Deaths

```r
# US-level: long format with metrics column
us_tidy <- us_df %>%
  pivot_longer(cols = c(cases, deaths),
               names_to  = "metrics",
               values_to = "count")

# US-level: per-capita metrics
us_tidy_per_pop <- us_df %>%
  mutate(
    cases_per_10k = cases  / population * 10000,
    deaths_per_1m = deaths / population * 1000000
  ) %>%
  pivot_longer(cols = c(cases_per_10k, deaths_per_1m),
               names_to  = "metrics_per_pop",
               values_to = "count_per_pop") %>%
  select(-c(cases, deaths))
```

```r
# State-level: long format with metrics column
states_tidy <- states_df %>%
  pivot_longer(cols = c(cases, deaths),
               names_to  = "metrics",
               values_to = "count")

# State-level: per-capita metrics
states_tidy_per_pop <- states_df %>%
  mutate(
    cases_per_10k = cases  / population * 10000,
    deaths_per_1m = deaths / population * 1000000
  ) %>%
  pivot_longer(cols = c(cases_per_10k, deaths_per_1m),
               names_to  = "metrics_per_pop",
               values_to = "count_per_pop") %>%
  select(-c(cases, deaths))
```

3.3.2 Daily New Cases & Deaths

```r
# US-level: compute daily new cases/deaths via lag differencing
# pmax(..., 0) clips negative values caused by retroactive data corrections
us_df <- us_df %>%
  arrange(date) %>%
  mutate(
    new_cases  = pmax(cases  - lag(cases,  default = 0), 0),
    new_deaths = pmax(deaths - lag(deaths, default = 0), 0)
  )

us_daily_tidy <- us_df %>%
  select(-c(cases, deaths)) %>%
```

```r
  pivot_longer(cols = c(new_cases, new_deaths),
               names_to  = "metrics",
               values_to = "count")
```

```r
# State-level: group by state before lag differencing to avoid cross-state bleed
# pmax(..., 0) clips negative values caused by retroactive data corrections
states_df <- states_df %>%
  group_by(state) %>%
  arrange(date) %>%
  mutate(
    new_cases  = pmax(cases  - lag(cases,  default = 0), 0),
    new_deaths = pmax(deaths - lag(deaths, default = 0), 0)
  ) %>%
  ungroup()

states_daily_tidy <- states_df %>%
  select(-c(cases, deaths)) %>%
  pivot_longer(cols = c(new_cases, new_deaths),
               names_to  = "metrics",
               values_to = "count")
```

---

4 Exploratory Data Analysis

4.1 Define Shared Plot Theme

```r
my_theme <- theme(
  plot.title  = element_text(face = "bold", size = 15, hjust = 0.5),
  axis.title  = element_text(face = "bold", size = 13),
  axis.text   = element_text(size = 10),
  legend.text = element_text(size = 10),
  legend.position = "bottom"
)
```

4.2 Overall U.S. Impact (Jan 2020 – Mar 2023)

4.2.1 Summary Statistics as of Mar 23, 2023

```r
us_df %>%
  filter(date == as.Date("2023-03-23")) %>%
  select(date, cases, deaths, population) %>%
  mutate(
    cases_per_10k      = cases  / population * 10000,
    deaths_per_10k     = deaths / population * 10000,
    case_fatality_rate = deaths / cases
  )
```

```
##         date     cases  deaths population cases_per_10k deaths_per_10k
## 1 2023-03-23 103910034 1135343  340380185      3052.764       33.35514
##   case_fatality_rate
## 1         0.01092621
```

Key findings:

- **Total cases**: ~103.9 million infected
- **Total deaths**: ~1.1 million deaths
- **Cases per 10k**: ~3,053 per 10,000 people
- **Deaths per 10k**: ~33.4 per 10,000 people
- **Case fatality rate**: ~1.1% of confirmed cases resulted in death

4.2.2 Top 10 States by Total Cumulative Impact

```r
states_tidy %>%
  filter(date == "2023-03-23") %>%
  group_by(metrics) %>%
  slice_max(order_by = count, n = 10) %>%
  ungroup() %>%
  mutate(state = reorder_within(state, count, metrics)) %>%
  ggplot(aes(x = state, y = count, fill = metrics)) +
  geom_col() +
  coord_flip() +
  scale_x_reordered() +
  facet_wrap(~metrics, scales = "free") +
  scale_fill_manual(
    name   = "Metrics",
    values = c("cases" = "royalblue", "deaths" = "tomato"),
    labels = c("cases" = "Cumulative Cases", "deaths" = "Cumulative Deaths")
  ) +
  labs(
    title = "Total Cases and Deaths by State (March 23, 2023)",
    x = "State", y = "Count"
  ) +
  theme_minimal() + my_theme
```

```r
ggsave("output/figures/fig1_top10_states_total.png", width = 10, height = 5, dpi = 300)
```

```r
cases_map <- plot_usmap(
  data    = states_tidy %>% filter(date == as.Date("2023-03-23"), metrics == "cases"),
  values  = "count",
  regions = "states",
  labels  = TRUE
) +
  scale_fill_continuous(name = "Cumulative Cases", low = "white", high = "royalblue",
                        label = scales::comma) +
  labs(title = "Cumulative Cases across the US (2023-03-23)") +
  theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
        legend.text = element_text(size = 7), legend.position = "right")
```

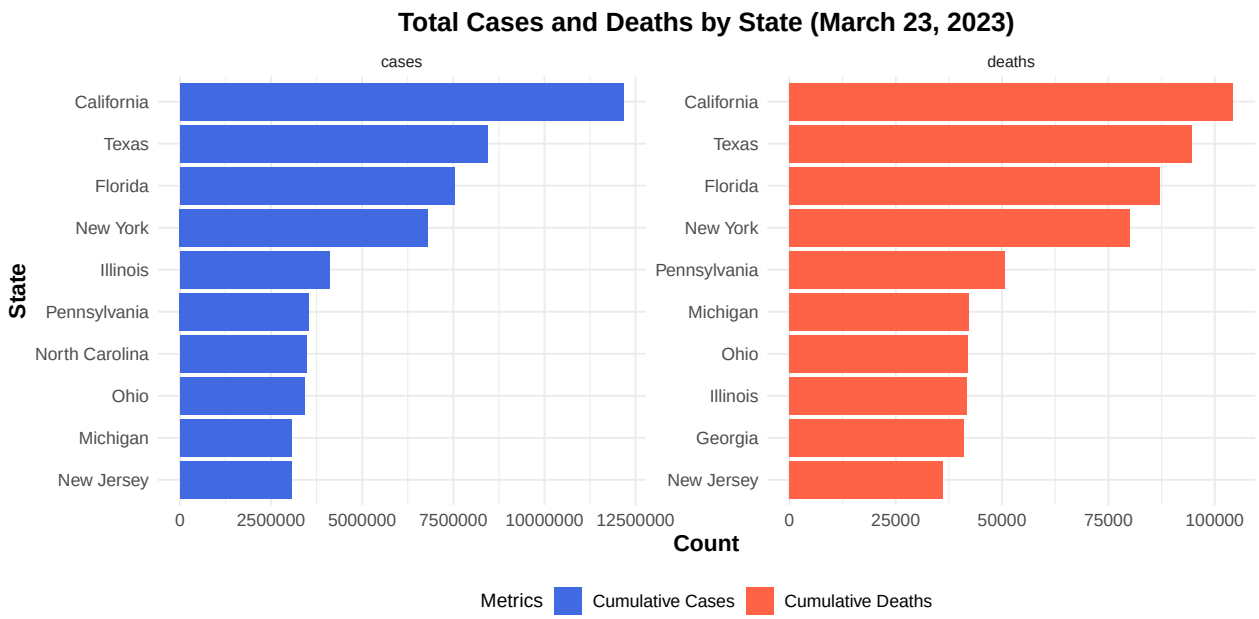

Figure 1: Top 10 states by cumulative cases and deaths (Mar 23, 2023).

```r
deaths_map <- plot_usmap(
  data    = states_tidy %>% filter(date == as.Date("2023-03-23"), metrics == "deaths"),
  values  = "count",
  regions = "states",
  labels  = TRUE
) +
  scale_fill_continuous(name = "Cumulative Deaths", low = "white", high = "tomato",
                        label = scales::comma) +
  labs(title = "Cumulative Deaths across the US (2023-03-23)") +
  theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
        legend.text = element_text(size = 7), legend.position = "right")

fig2 <- cases_map | deaths_map
fig2
```

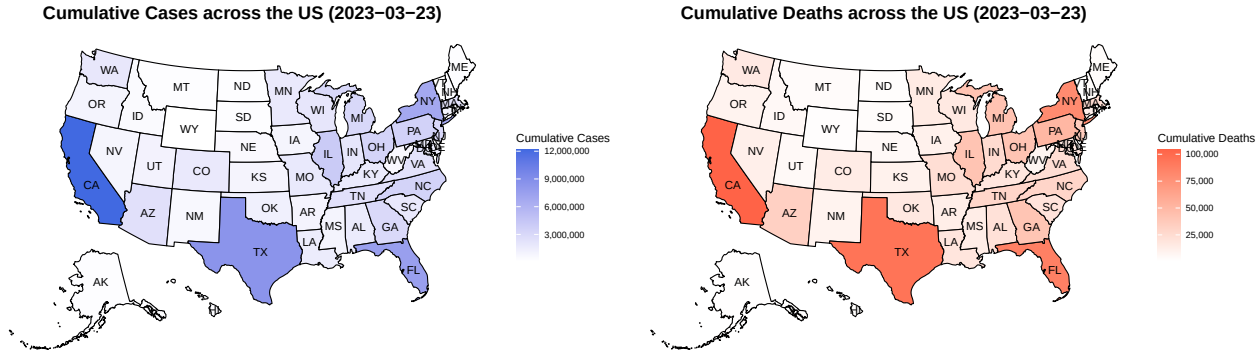

Figure 2: Geographic distribution of cumulative cases and deaths across U.S. states (Mar 23, 2023).

```r
ggsave("output/figures/fig2_maps_cumulative_total.png", plot = fig2, width = 14, height = 8, dpi = 300)
```

Findings: California, Texas, Florida, and New York had the highest total cases and deaths — likely driven by their large population bases.

4.2.3 Top 10 States by Per-Capita Impact

```r
states_tidy_per_pop %>%
  filter(date == "2023-03-23") %>%
  group_by(metrics_per_pop) %>%
  slice_max(order_by = count_per_pop, n = 10) %>%
  ungroup() %>%
  mutate(state = reorder_within(state, count_per_pop, metrics_per_pop)) %>%
  ggplot(aes(x = state, y = count_per_pop, fill = metrics_per_pop)) +
  geom_col() +
  coord_flip() +
  scale_x_reordered() +
  facet_wrap(~metrics_per_pop, scales = "free") +
  scale_fill_manual(
    name   = "Per-Capita Metrics",
    values = c("cases_per_10k" = "royalblue", "deaths_per_1m" = "tomato"),
    labels = c("cases_per_10k" = "Cases per 10k", "deaths_per_1m" = "Deaths per 1m")
  ) +
  labs(
    title = "Per-Capita Cases and Deaths by State (March 23, 2023)",
    x = "State", y = "Count"
  ) +
  theme_minimal() + my_theme
```

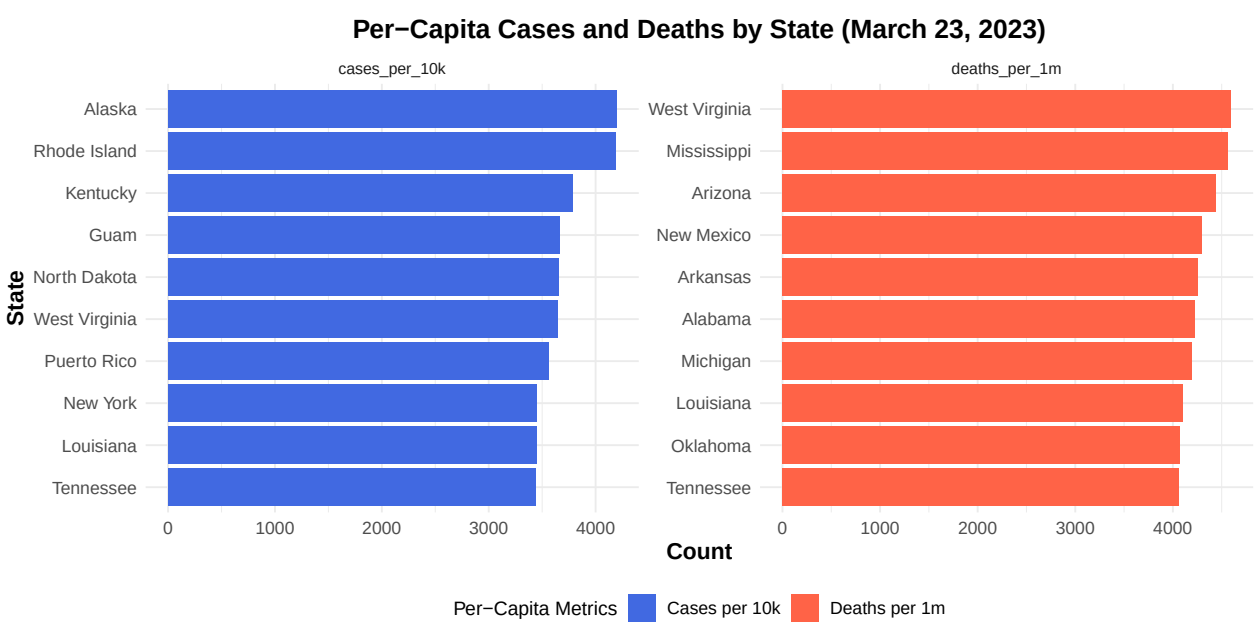

Figure 3: Top 10 states by per-capita cumulative cases and deaths (Mar 23, 2023).

```r
ggsave("output/figures/fig3_top10_states_percapita.png", width = 10, height = 5, dpi = 300)
```

```r
cases_per_map <- plot_usmap(
  data    = states_tidy_per_pop %>%
              filter(date == as.Date("2023-03-23"), metrics_per_pop == "cases_per_10k"),
  values  = "count_per_pop",
  regions = "states",
  labels  = TRUE
) +
  scale_fill_continuous(name = "Cases per 10k", low = "white", high = "royalblue",
                        label = scales::comma) +
  labs(title = "Cumulative Cases per 10k People (2023-03-23)") +
  theme(plot.title = element_text(face = "bold", size = 10, hjust = 0.5),
        legend.text = element_text(size = 7), legend.position = "right")

deaths_per_map <- plot_usmap(
  data    = states_tidy_per_pop %>%
              filter(date == as.Date("2023-03-23"), metrics_per_pop == "deaths_per_1m"),
  values  = "count_per_pop",
  regions = "states",
  labels  = TRUE
) +
  scale_fill_continuous(name = "Deaths per 1m", low = "white", high = "tomato",
                        label = scales::comma) +
  labs(title = "Cumulative Deaths per 1m People (2023-03-23)") +
  theme(plot.title = element_text(face = "bold", size = 10, hjust = 0.5),
        legend.text = element_text(size = 7), legend.position = "right")

fig4 <- cases_per_map | deaths_per_map
fig4
```

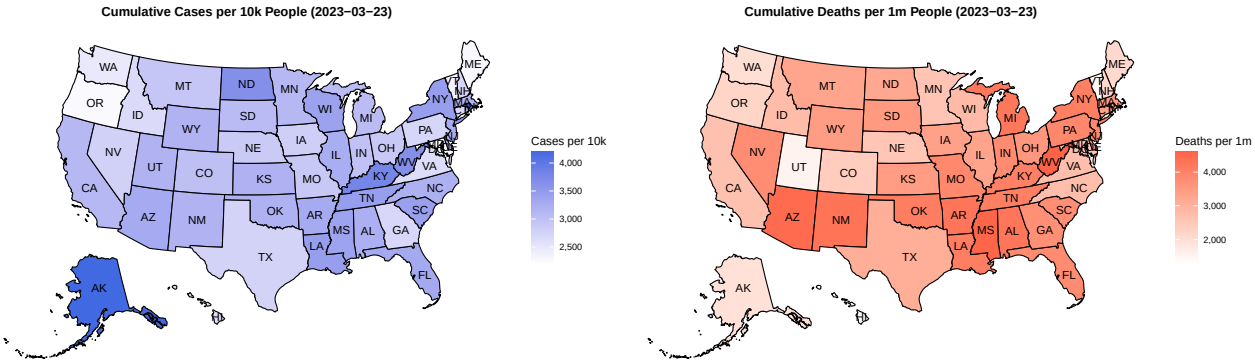

Figure 4: Geographic distribution of cumulative cases and deaths across U.S. states (Mar 23, 2023).

```r
ggsave("output/figures/fig4_maps_cumulative_percapita.png", plot = fig4, width = 14, height = 8, dpi = 3
```

Findings: Per-capita analysis reveals a different picture — **Alaska and Rhode Island** top cases (>4,000 per 10k), while **West Virginia, Mississippi, and Arizona** lead in deaths (>4,500 per 1M).

4.3 Temporal Evolution & Seasonality

4.3.1 Cumulative Trend with Variant Milestones

```r
# Key pandemic events for annotation
events <- data.frame(
  event_date = as.Date(c("2020-03-11", "2020-12-29", "2021-07-01", "2021-12-01")),
  label      = c("WHO Declared Pandemic", "Alpha Variant", "Delta Variant", "Omicron Variant")
)

ggplot(us_tidy_per_pop) +
  geom_line(aes(x = date, y = count_per_pop, color = metrics_per_pop,
                linetype = metrics_per_pop), linewidth = 1) +
  scale_color_manual(
    name   = "Metrics",
    values = c("cases_per_10k" = "royalblue", "deaths_per_1m" = "tomato"),
    labels = c("cases_per_10k" = "Cases per 10k", "deaths_per_1m" = "Deaths per 1m")
  ) +
  scale_linetype_manual(
    name   = "Metrics",
    values = c("cases_per_10k" = "solid", "deaths_per_1m" = "dashed"),
    labels = c("cases_per_10k" = "Cases per 10k", "deaths_per_1m" = "Deaths per 1m")
  ) +
  scale_y_continuous(
    name     = "Cases per 10k",
    n.breaks = 10,
    labels   = scales::comma,
    sec.axis = sec_axis(~., name = "Deaths per 1m", labels = scales::comma)
  ) +
  scale_x_date(name = "Date", date_labels = "%Y-%m", date_breaks = "2 months") +
  geom_vline(data = events, aes(xintercept = event_date),
             linetype = "dashed", color = "grey40") +
  geom_text(data = events, aes(x = event_date, y = Inf, label = label),
            angle = 90, vjust = 1.5, hjust = 1.1, size = 3, color = "grey30") +
  labs(
    title    = "COVID-19 Cumulative Cases and Deaths Per Capita in the US",
    subtitle = "(Jan 2020 - Mar 2023)"
  ) +
  theme_minimal() + my_theme +
  theme(plot.subtitle = element_text(hjust = 0.5),
        axis.text.x   = element_text(angle = 45, hjust = 1))
```

```r
ggsave("output/figures/fig5_cumulative_trend_percapita.png", width = 10, height = 5, dpi = 300)
```

Findings: Five distinct pandemic phases are visible: (1) initial outbreak, (2) winter surge 2020–21, (3) Delta wave summer 2021, (4) Omicron surge winter 2021–22, and (5) post-pandemic stabilization. The death curve notably lags cases and does not spike as sharply during Omicron, suggesting vaccination effects.

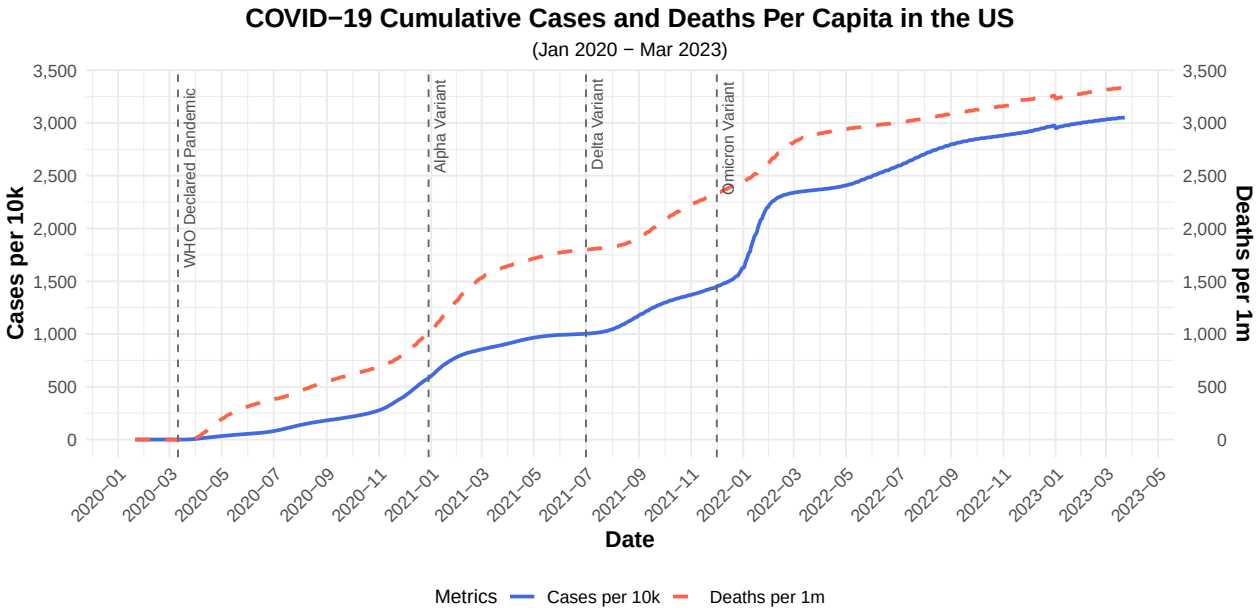

Figure 5: Cumulative COVID-19 cases and deaths in the U.S. (Jan 2020 – Mar 2023), with key milestones annotated.

4.3.2 Monthly New Cases & Deaths

```
us_monthly <- us_daily_tidy %>%
  mutate(month_start = floor_date(date, unit = "month")) %>%
  group_by(month_start, metrics) %>%
  summarise(monthly_total = sum(count, na.rm = TRUE), .groups = "drop")

ggplot(us_monthly, aes(x = month_start, y = monthly_total, color = metrics)) +
  geom_line(linewidth = 1) +
  geom_point(size = 1) +
  scale_color_manual(
    name   = "Metrics",
    values = c("new_cases" = "royalblue", "new_deaths" = "tomato"),
    labels = c("new_cases" = "New Cases", "new_deaths" = "New Deaths")
  ) +
  facet_wrap(~metrics, scales = "free_y", ncol = 1) +
  scale_y_continuous(labels = scales::comma) +
  scale_x_date(date_labels = "%Y-%m", date_breaks = "2 months") +
  labs(
    title    = "Monthly New COVID-19 Cases and Deaths in the US",
    subtitle = "(Jan 2020 - Mar 2023)",
    x = "Month", y = "Monthly Count"
  ) +
  theme_minimal() + my_theme +
  theme(plot.subtitle = element_text(hjust = 0.5),
        axis.text.x   = element_text(angle = 45, hjust = 1))
```

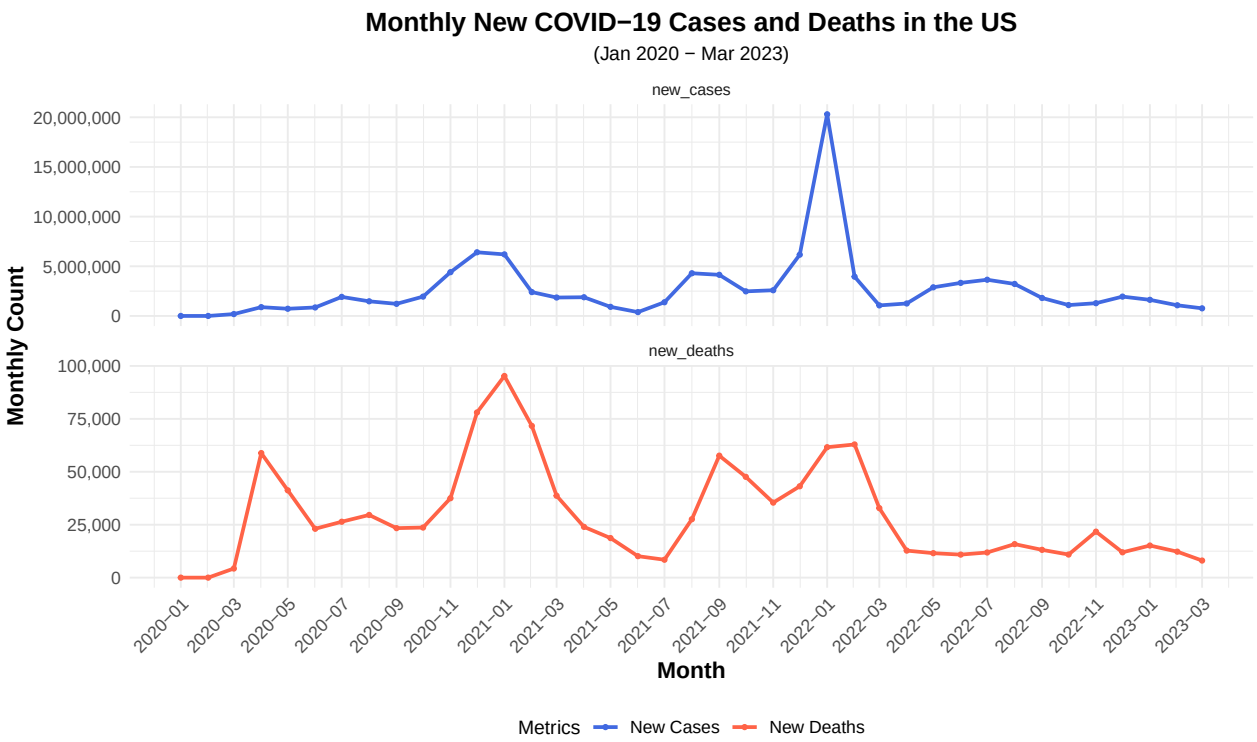

Figure 6: Monthly totals of new COVID-19 cases and deaths in the U.S. (Jan 2020 – Mar 2023).

```
ggsave("output/figures/fig6_monthly_new_cases_deaths.png", width = 10, height = 6, dpi = 300)
```

Findings: Largest death peak: Jan 2021 (first winter surge). Largest case peak: Jan 2022 (Omicron). The lowest values in both occur around Jul 2021, likely reflecting vaccination rollout and herd immunity effects.

4.4 Geographic Distribution Analysis

4.4.1 Severely Impacted States

```
severely_impacted <- c("Alaska", "Rhode Island", "West Virginia", "Mississippi",
                       "Arizona", "California", "Texas", "Florida", "New York")

map_states <- data.frame(state = severely_impacted, group = "Severely Impacted")

plot_usmap(data = map_states, values = "group", labels = TRUE) +
  scale_fill_manual(
    name   = "Category",
    values = c("Severely Impacted" = "orange"),
    labels = c("Severely Impacted", "Other States"),
    na.value = "gray90"
  ) +
```

```
  labs(title = "Severely Impacted US States") +
  theme(plot.title = element_text(face = "bold", size = 10, hjust = 0.5),
        legend.text = element_text(size = 7), legend.position = "right")
```

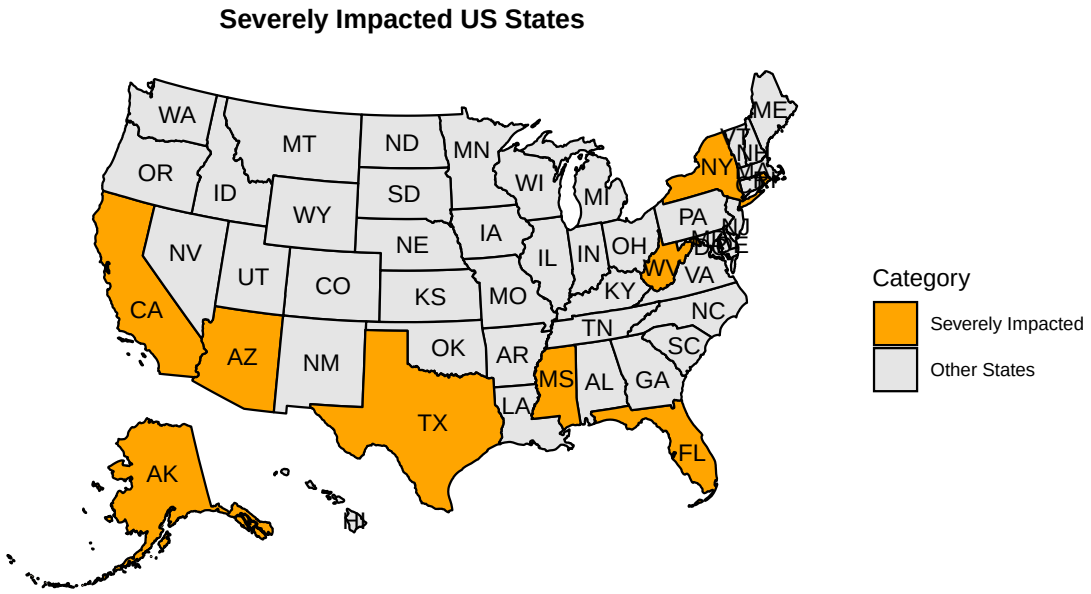

Figure 7: U.S. states most severely impacted by COVID-19, identified by both total and per-capita burden.

```
ggsave("output/figures/fig7_map_severely_impacted_states.png", width = 6, height = 6, dpi = 300)
```

4.4.2 State-Level 30-Day Rolling Average

```
states_daily_subset <- states_daily_tidy %>%
  filter(state %in% severely_impacted) %>%
  mutate(metrics = factor(metrics, levels = c("new_cases", "new_deaths"))) %>%
  group_by(state, metrics) %>%
  arrange(date) %>%
  mutate(rolling_count = rollmean(count, k = 30, fill = NA, align = "right")) %>%
  ungroup()

fig8 <- states_daily_subset %>%
  filter(!is.na(rolling_count)) %>%
  ggplot(aes(x = date, y = rolling_count, color = metrics)) +
  geom_line(linewidth = 1) +
  scale_color_manual(
    name   = "Metrics",
    values = c("new_cases" = "royalblue", "new_deaths" = "tomato")
  ) +
  facet_wrap(state ~ metrics, scales = "free", ncol = 2) +
  scale_x_date(date_labels = "%Y-%m", date_breaks = "4 months") +
  labs(
    title = "30-Day Rolling Average of New Cases vs. New Deaths",
    x = "Date", y = "30-Day Rolling Average"
```

```
  ) +
  theme_minimal() +
  theme(
    plot.title    = element_text(face = "bold", size = 15, hjust = 0.5),
    strip.text    = element_text(face = "bold", size = 13),
    axis.text.x   = element_text(size = 10, angle = 45, hjust = 1),
    legend.position = "none"
  )

fig8
```

```
ggsave("output/figures/fig8_rolling_avg_impacted_states.png", plot = fig8, width = 15, height = 20, dpi
```

Key findings:

1. **Omicron Spike (Jan 2022):** Cases spiked massively nationwide; deaths rose more modestly, especially in CA and NY — suggesting vaccine protection against severe outcomes.
2. **Regional Seasonality:** Southern states (AZ, FL, MS, TX) show summer waves; northern states (NY, RI) peak in winter.
3. **Decoupling Over Time:** Post-2022, case counts stay lower and deaths drop further — reflecting vaccination, immunity, and milder variants.
4. **NY Early Anomaly:** New York shows high early deaths with low confirmed cases, likely due to limited testing in early 2020.

---

5 Summary

This project reconstructs the U.S. COVID-19 pandemic trajectory from **January 21, 2020 to March 23, 2023**, integrating multi-source data to evaluate public health burden, temporal evolution, and geographic heterogeneity.

Key findings:

- A total of **103.9 million cases** and **1.1 million deaths** were recorded nationally.
- The pandemic progressed through **five distinct waves**, with the Omicron surge (Jan 2022) producing the highest case counts.
- A clear **decoupling of cases and deaths** emerged during Omicron, likely driven by vaccination and natural immunity.
- **Regional seasonality** was evident: southern states peaked in summer; northern states peaked in winter.
- Per-capita analysis highlighted disproportionate impact in **smaller states** (Alaska, Rhode Island, West Virginia).

Limitations:

- Early testing shortages undercount infections, particularly in New York in early 2020.
- External vaccination coverage data were not integrated — limiting causal inference about fatality rate decline.
- Analysis ends in March 2023 and does not capture subsequent variant developments.

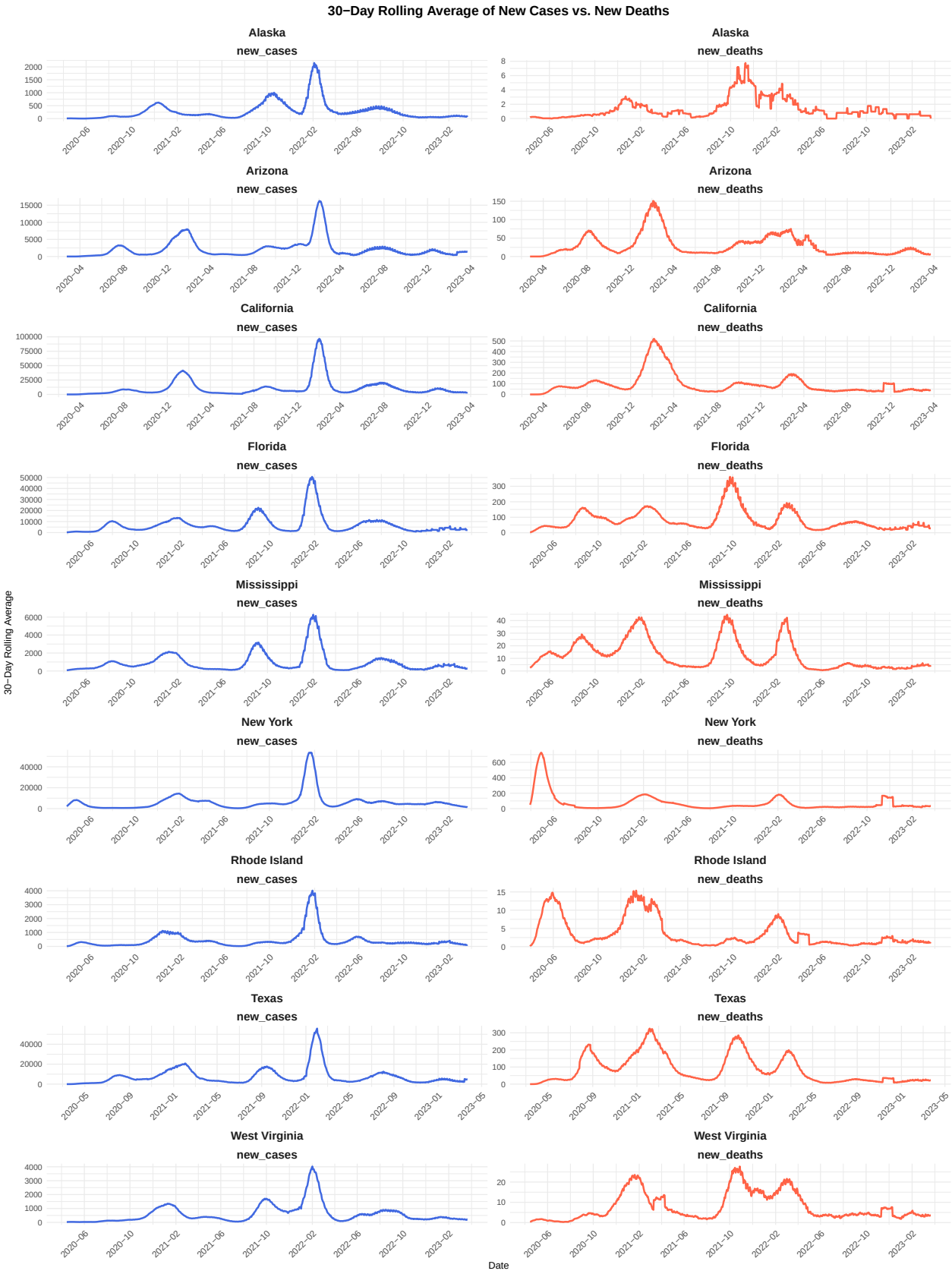

Figure 8: 30-day rolling average of daily new cases and deaths in the nine severely impacted states (Jan 2020 – Mar 2023).